

Lab03 Introduction to Python

Assignments:

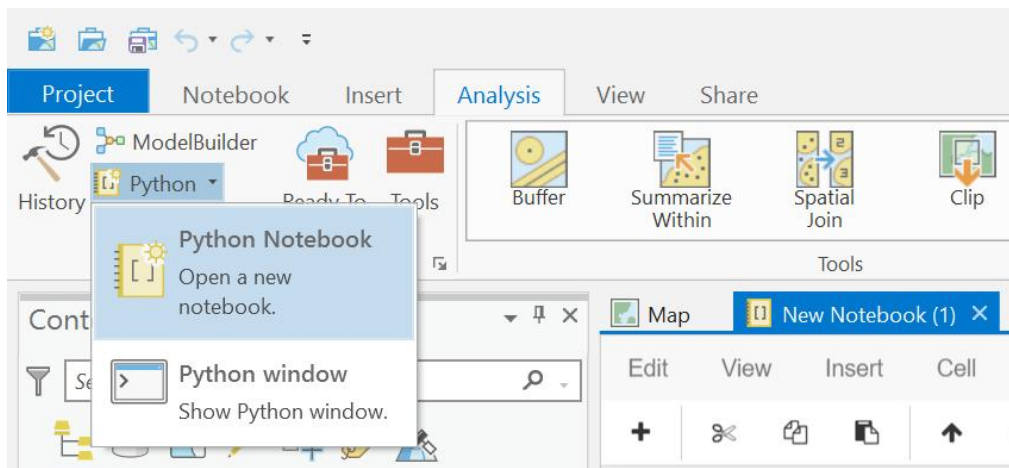
1. Fill in all the blanks in the following exercises;
2. Attach the screen shots of Task 1 (capture the attribute table and the map for the updated “streetname”) and Task 2 (capture the attribute table for “NewStrLen”);
3. Attach python scripts for Exercise 7 (either with .py file or attach your scripts in the word file)

Open the ArcGIS pro

Start a new project

Please copy the data to your owner folder

Start the Python notebook



1. Other noticed in Python script

1.1. Comments

Comments in Python start with the hash character, #, and extend to the end of the physical line. Since comments are to clarify code and are not interpreted by Python, they may be omitted when typing in examples. Write comments is a good habit for excellent programmer when writing Python script. Readers can be easily understanding the meaning of steps.

```
# this is a comment and it can be omitted
```

1.2. Indentation

Statement grouping is done by indentation instead of beginning and ending brackets in python

1.3. Case Sensitive

“PRINT” and “print” is different, try it later.

2. Variable operations

2.1. String operations

2.1.1. Assign variable to string

String can be enclosed in single quotes ('...') or double quotes ("...") with the same result. \ can be used to escape quotes.
How to assign a variable to store string? = I am Peter Chan, I'm studying in HKU.

```
>>> name = "I am Peter Chan"
```

```
>>> print(name)
```

```
_I am Peter Chan_
```

```
>>> name2 = 'I'm Peter Chan.'
```

```
>>> print(name2)
```

```
_I'm Peter Chan._
```

```
>>> name3 = 'I'm Peter Chan. \nI am studying in HKU'
```

```
>>> print(name3)
```

```
I'm Peter Chan.
```

```
I am studying in HKU.
```

Q: Try "PRINT (name)" and "print (NAME)" to see how case sensitive means.

It'll result in error, because Python only accepts lowercase 'print' and the uppercase 'NAME' does not exist.

2.1.2. String Concatenation

String concatenation

```
>>> n = "I'm Peter Chan" + "I am studying in HKU"
```

```
>>> print(n)
```

```
I'm Peter ChanI am studying in HKU
```

```
>>> n1 = "I'm Peter Chan"
```

```
>>> n2 = "I am studying in HKU"
```

```
>>> n3 = n1 + " " + n2
```

```
>>> print(n3)
```

```
I'm Peter Chan I am studying in HKU
```

String multiplication and concatenation

```
>>> n4 = n1 * 2 + "\n" + n2
```

```
>>> print(n4)
```

I'm Peter ChanI'm Peter Chan
I am studing in HKU

2.1.3. String Indexing

String index

```
>>> n1 = "Hello world"  
>>> n1[0]
```

H

```
.>>>n1[1]
```

e

```
>>>n1[-1]
```

d

2.1.4. String Slicing

String slicing

```
>>> n1 = "Hello world"  
>>> n1[0:4]
```

Q: The index is started from number 0? If you want to slice the “Hello wo”, you will input n1[0:7]? No, it should be n1[0:8]
In the script n1[0:4], is n1[4] included? No

String len function

```
>>> n1 = "Hello world"  
>>> len(n1)  
11
```

2.2. Lists operations

Lists can contain items of different types, but usually the items all have the same type as numbers or strings.

```
>>> s = [1,4,9,16,25]  
>>> print (s)  
[1, 4, 9, 16,25]  
.>>> print (s[1]) #list indexing, start with 0  
4  
>>> print (s[-1]) #last index is -1  
25
```

```

>>> print(s[2:4]) #slicing
__9, 16__

.>>> w = ["Apple", "Banana", "Orange"]
>>> print(w[0])
__Apple__

>>> nw= w + ['a', 'b', 'c'] #slice concatenation
>>> print(nw)
__['Apple', 'Banna', 'Orange', 'a', 'b', 'c']__

>>> nw[2:4]= [1,2] #assign to slices
>>> print(nw)
__['Apple', 'Banna', 1, 2, 'b', 'c']__

>>> len(nw)
__6__

```

2.3. Number operations

2.3.1 Calculator +,-,*,/,%,**

```

>>> 17 / 3 # int / int -> int or float?
__5.666666666666667__

>>> 17 % 3 # the % operator returns the remainder of the division
__2__

>>> 5 * 3 + 2 - 1 # result * divisor + remainder - int
__16__

>>> 5 ** 2 # 5 squared
__25__

>>> 2 ** 7 # 2 to the power of 7
__128__

```

Q: What is the difference between “/” and “%” in Python script?

The '/' operator performs division and returns the quotient (usually a floating-point number with a decimal). The '%' operator performs modulo operation and returns only the remainder after division.

2.3.2 Assign numbers

The equal sign (=) is used to assign a value to a variable.

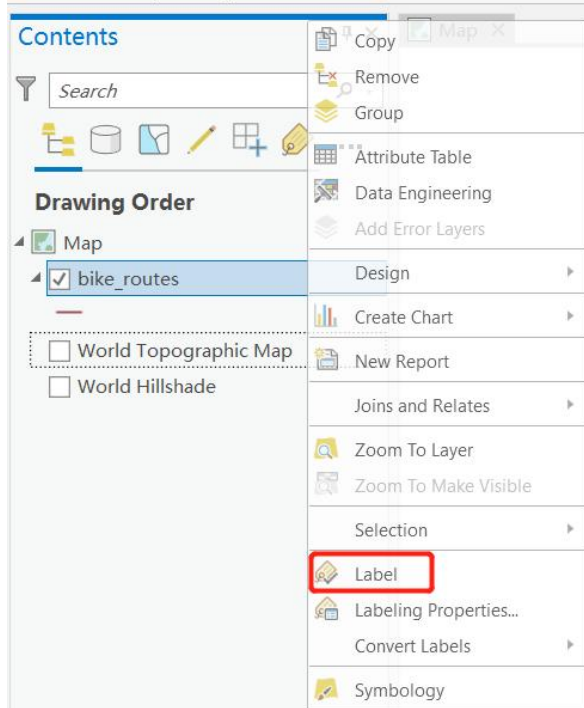
```

>>> width = 20
>>> height = 5 * 9
>>> area = width * height
>>> area

```

Task1: Add your name in the end of the street name in the attribute table.

1. Right click the bike_routes layer.
2. Enable the label and you can double click the any row to zoom to the feature.



3. Click the labeling properties.

The screenshot shows the QGIS interface with a map of San Francisco. The 'Labeling Properties...' dialog box is open, and the 'Label' tab is selected. The dialog box is highlighted with a red rectangle. Below the map, a table of street data is visible.

Shape	objectid	cnn	streetname	from_st	to_st
Polyline	3421	8012000	LAKE ST	4TH AV	5TH AV
Polyline	3567	8020000	LAKE ST	12TH AV	FUNSTON AV
Polyline	9620	2322000	ANZA ST	6TH AVE	5TH AVE
Polyline	1401	8016000	LAKE ST	8TH AV	9TH AV
Polyline	9619	2321000	ANZA ST	5TH AVE	4TH AVE
Polyline	5005	5123000	EL CAMINO DEL MAR	27TH AV	28TH AV
Polyline	9608	2310000	ANZA ST	SPRUCE ST	COOK ST
Polyline	2008	8018000	LAKE ST	10TH AV	11TH AV
Polyline	4033	12742000	TURK ST	ANNAPOLIS TERR	TAMALPAIS TERR
Polyline	2190	5126000	EL CAMINO DEL MAR	30TH AV	LAKE ST
Polyline	2851	12751000	TURK ST	ROSSI AV	WILLARD NORTH ST
Polyline	2384	6286000	GOLDEN GATE AVE	CHABOT TER	TEMASCAL TER
Polyline	3529	8848000	MASONIC AVE	MC ALLISTER ST	FULTON ST

4. Choose the Language “Python”
5. Input the code to combine the strings as the following picture. (Use your own name),and then check the label of the road name.

Label Class - bike_routes

Class 1

Class Symbol Position



Language Python

Fields

FID

objectid

cnn

streetname

from_st

to_st

facility_t

direct

surface tr

Insert Values

Functions

.as_integer_ratio()

.capitalize()

.center()

.conjugate()

.count()

.decode()

.denominator()

.encode()

.endswith()

Expression

☐ Advanced

[streetname]+'JHe'



☒ Use coded value descriptions

☒ Remove extra spaces

☐ Remove extra line breaks

[Learn more about label expressions](#)

Apply

Cancel

Catalog Geoprocessing Symbology Label Class

Task2: Recalculate the street length by street length divided by 1000.

Open the attribute table and add a new field and name it as NewStrLen.

bike_routes *Fields: bike_routes X

Current Layer: bike_routes

	Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Default	Precision	Scale	Length
	☒	☐	buffered	buffered	Text	☐	☐			0	0	254
	☒	☐	raised	raised	Text	☐	☐			0	0	254
	☒	☐	contraflow	contraflow	Text	☐	☐			0	0	254
	☒	☐	greenwave	greenwave	Text	☐	☐			0	0	254
	☒	☐	biap	biap	Text	☐	☐			0	0	254
	☒	☐	sm_sweeper	sm_sweeper	Text	☐	☐			0	0	254
	☒	☐	recommenda	recommenda	Text	☐	☐			0	0	254
	☒	☐	date_data_	date_data_	Date	☐	☐			0	0	
	☒	☐	time_data_	time_data_	Text	☐	☐			0	0	254
	☒	☐	date_dat_2	date_dat_2	Text	☐	☐			0	0	
	☒	☐	time_dat_2	time_dat_2	Text	☐	☐			0	0	254
	☒	☐	NewStrLen		Double	☐	☐					

Click Calculate and input the expression.

bike_routes

*Fields: bike_routes

Field:

Add

Calculate

Selection:

Select By Attributes

Zoom To

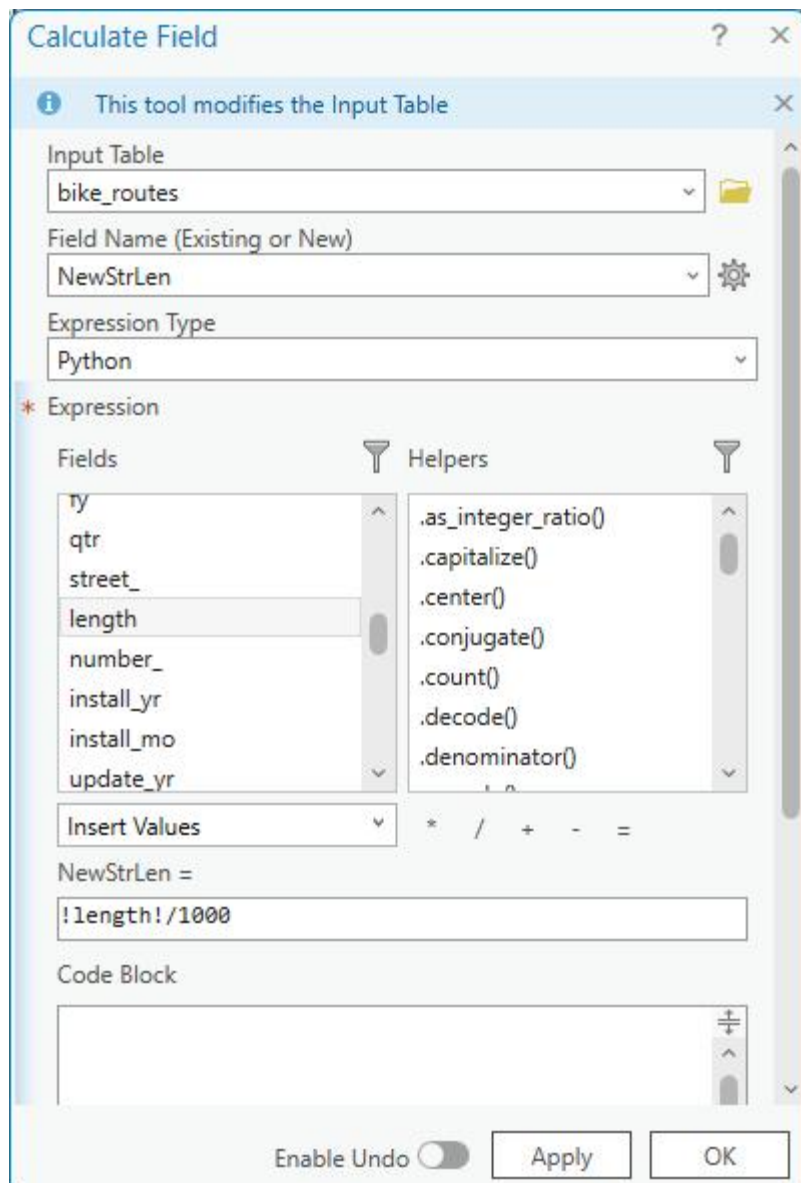
Switch

Clear

Delete

Copy

	FID	Shape	objectid	cnn	streetname	from_st	to_st
1	0	Polyline	3421	8012000	LAKE ST	4TH AV	5TH AV
2	1	Polyline	3567	8020000	LAKE ST	12TH AV	FUNSTON AV
3	2	Polyline	9620	2322000	ANZA ST	6TH AVE	5TH AVE
4	3	Polyline	1401	8016000	LAKE ST	8TH AV	9TH AV
5	4	Polyline	9619	2321000	ANZA ST	5TH AVE	4TH AVE
6	5	Polyline	5005	5123000	EL CAMINO DEL MAR	27TH AV	28TH AV
7	6	Polyline	9608	2310000	ANZA ST	SPRUCE ST	COOK ST
8	7	Polyline	2008	8018000	LAKE ST	10TH AV	11TH AV
9	8	Polyline	4033	12742000	TURK ST	ANNAPOLIS TERR	TAMALPAIS TERR
10	9	Polyline	2190	5126000	EL CAMINO DEL MAR	30TH AV	LAKE ST
11	10	Polyline	2851	12751000	TURK ST	ROSSI AV	WILLARD NORTH ST
12	11	Polyline	2384	6286000	GOLDEN GATE AVE	CHABOT TER	TEMASCAL TER
13	12	Polyline	3529	8848000	MASONIC AVE	MC ALLISTER ST	FILTON ST



3 Functions and methods

3.1 Function and method on list

Difference between method and function

Simply, function and method both look similar as they perform in almost similar way.

But method is closely coupled with an object.

Functions can be called only by its name, as it is defined independently. But methods can't be called by its name only, we need to instantiate an object to call the properties and methods of the class, method is defined within a class and hence they are dependent on that class.

```
>>> stack = [1,2,3,4,5]
```

```
>>> stack.append(6) #add 6 into the list
```

```
>>> stack
_____ [1, 2, 3, 4, 5, 6] _____
>>> stack.pop() #pop out last item in the list
>>> stack
_____ [1, 2, 3, 4] _____
```

```
List.sort(), List.count()
.>>> num = [13, 4, 6, 3, 18, 9, 25, 3]
>>> num.sort ()
>>> num
_____ [3, 3, 4, 6, 9, 13, 18, 25] _____
.>>> num.count(3) #counting how many '3' in the list.
_____ 2 _____
```

```
list del
>>> a = [-1, 1, 66.25, 333, 333, 1234.5]
>>> del a[0]
>>> a
_____ [1, 66.25, 333, 333, 1234.5] _____
>>> del a[2:4]
>>> a
_____ [1, 66.25, 1234.5] _____

Built in function
>>> d = pow (2,4)
>>> print (d)
_____ 16 _____
```

4.Modules and Packages

A module is a file containing Python function definitions and statements. The file name is the module name with the suffix .py appended.

A Python package usually consists of several modules. Physically, a package is a folder containing modules and maybe other folders that themselves may contain more folders and modules. Conceptually, it's a namespace. This simply means that a package's modules are bound together by a package name, by which they may be referenced.

Build in module
<pre>import math print (dir (math))</pre>
Results:['__doc__', '__loader__', '__name__', '__package__', '__spec__', 'acos', 'acosh', 'asin', 'asinh', 'atan', '']

5.Class

```
class Dog:
    kind = 'canine'
    def run(self):
        print('I can run very fast!')

    def eat(self):
        print('Bones are my favourite food!')
```

```
>>> d = Dog()
```

```
>>> d.kind
```

```
_____ 'canine' _____
```

```
>>> d.run()
```

```
_____ I can run very fast! _____
```

6.Control flow tools: conditional statement (if, while, for)

a. “If” statement

Lab01_if01.py
<pre>import random p= random.randint(1,6) print (p) if p ==6: print ("You win") elif p ==5: print ("Try again!") else: print ("You lose!")</pre>

Lab01_if02.py (with user input)
<pre>import random p= random.randint(1,6) guess= int(input("Please input an integer between 1 – 6: ")) if guess == p: print ("You win")</pre>

<pre>else: print ("You lose!")</pre>
Q7: What would happen if no space before “print (“You win”)? _An IndentationError occurs when Python uses indentation to identify code blocks. The lack of indentation prevents the program from determining if a statement belongs to an if branch.

b. “While” statement

Lab01_while01.py
<pre>i=0 s=0 while i <=10: print (i) s = s+ i i= i+1 print ("sum of 1 to 10 = ", s)</pre>
Result: sum of 1 to 10 = 55

c. “For” statement

Lab01_for01.py
<pre>mylist = [1,2,3,4,5] for n in mylist : print (n)</pre>
Result: 1 2 3 4 5

Lab01_for02.py
<pre>myrange = range(0,101, 10) for n in myrange : print (n)</pre>
Result: 0 10 20 30 40 50 60 70 80 90

range(stop) -> range object
range(start, stop[, step]) -> range object

Return an object that produces a sequence of integers from start (inclusive) to stop (exclusive) by step. range(i, j) produces i, i+1, i+2, ..., j-1. start defaults to 0, and stop is omitted! range(4) produces 0, 1, 2, 3. These are exactly the valid indices for a list of 4 elements. When step is given, it specifies the increment (or decrement).

`range()`

.Lab01 for03.py

```
a = ['Mary', 'had', 'a', 'little', 'lamb']
for i in range(len(a)):
    print(i, a[i])
```

Result:
0 Mary
1 had
2 a
3 little
4 lamb

d. Break Statement, and else clauses on loops

The break statement, like in C, breaks out of the smallest enclosing “for” or “while” loop. Loop statements may have an *else* clause; it is executed when the loop terminates through exhaustion of the list (with for) or when the condition becomes false (with while), but not executed when the loop is terminated by a break statement.

```
fruit = ["apple", "banana", "orange", "grape", "pear"]
g = "orange"
for i in fruit:
    if i == g :
        print(g, " is in my fruit list")
        break
else :
    print("cannot find ",g, " in fruit list")
```

Result:
orange is in my fruit list

7. Exercise

Ex01: Create for loop to calculate the sum of $(1+2+3+\dots+99)$.

```
total_sum = 0
for i in range(1, 100):
    total_sum += i
print("Sum of 1 to 99 =", total_sum)
```

Ex02: Create a script that examines a string for the occurrence of a particular letter. If letter occurs in the text, print “Yes”, else print “No”.

```
text = "Hello HKU"
letter = "H"
if letter in text:
    print("Yes")
else:
    print("No")
```

Ex03: Create a script that examines a list of numbers without duplicates (for examples, 2, 8, 64, 16, 32, 4), and determines the second-largest number. Tips: use list sort function.

```
num_list = [2, 8, 64, 16, 32, 4]
num_list.sort()
print("The second-largest number is:", num_list[-2])
```

Ex04: Create a script that examines a list of numbers with duplicate values, remove the duplicates from the list, and print out non-duplicate list.

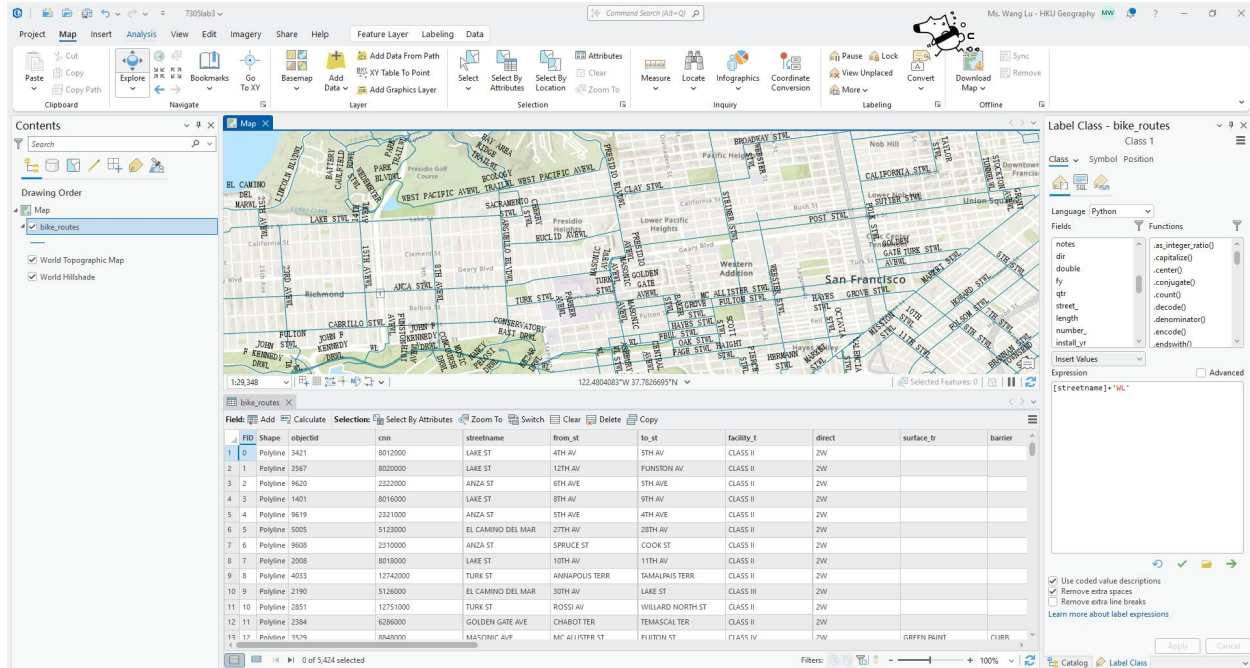
```
duplicates = [1, 2, 2, 3, 4, 4, 5]
non_duplicates = list(set(duplicates))
print("Non-duplicate list:", non_duplicates)
```

8. Reference

Python tutorial

<http://docs.python.org/tutorial/>

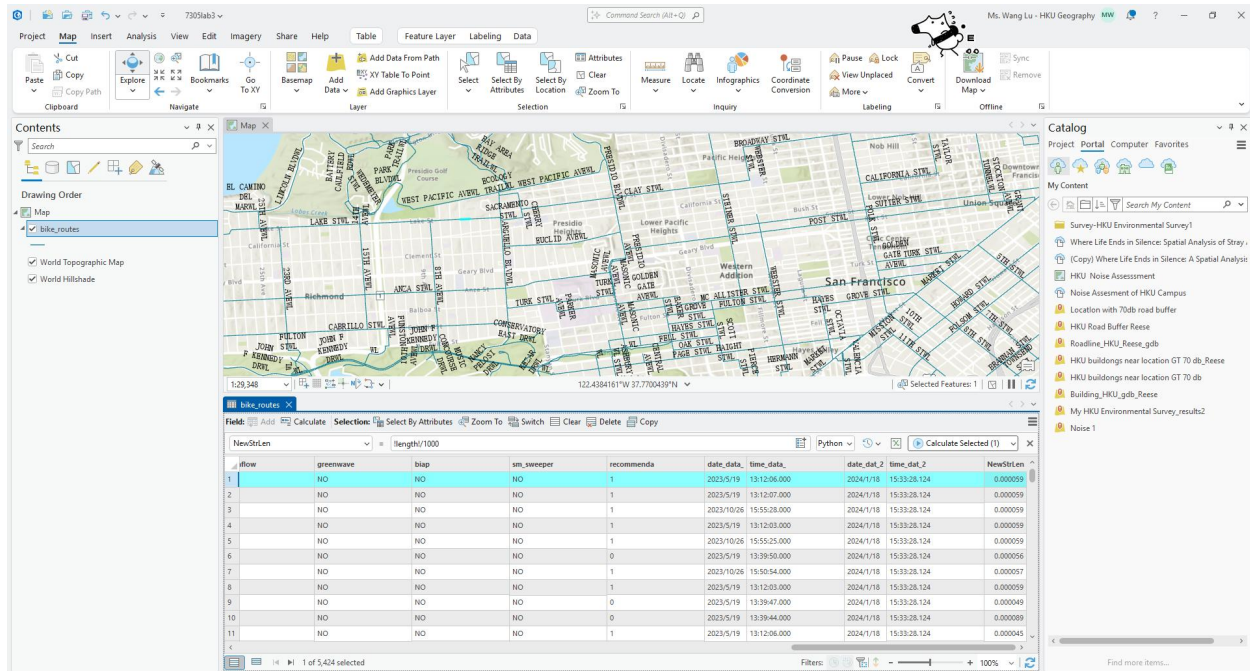
Task1



The screenshot shows the QGIS 3.16.0 interface. The main map displays a street network in San Francisco. The 'bike_routes' layer is selected in the 'Contents' panel. The 'Label Class - bike_routes' panel is open on the right, showing a list of functions and a Python expression field. The expression field contains the text `[streetname]>'ML'`. The 'Fields' panel at the bottom shows the attributes of the selected features, including 'id', 'shape', 'objid', 'cns', 'streetname', 'from_id', 'to_id', 'direct', 'surface', and 'barrier'.

ID	Shape	objid	cns	streetname	from_id	to_id	direct	surface	barrier
1	Polyline	3421	8012000	LAKE ST	47th AV	5TH AV	CLASS II	2W	
2	Polyline	3567	8023000	LAKE ST	12TH AV	FULTON AV	CLASS II	2W	
3	Polyline	3620	2322000	ANZA ST	6TH AV	5TH AV	CLASS II	2W	
4	Polyline	1401	8016000	LAKE ST	8TH AV	9TH AV	CLASS II	2W	
5	Polyline	9619	2321000	ANZA ST	5TH AV	4TH AV	CLASS II	2W	
6	Polyline	5005	5123000	EL CAMINO DEL MAR	27TH AV	28TH AV	CLASS II	2W	
7	Polyline	9608	2310000	ANZA ST	SPRUCE ST	COOK ST	CLASS II	2W	
8	Polyline	2008	8018000	LAKE ST	10TH AV	11TH AV	CLASS II	2W	
9	Polyline	4033	12742000	TURK ST	ANNAPOLIS TERR	TAMALPAIS TERR	CLASS II	2W	
10	Polyline	2190	5128000	EL CAMINO DEL MAR	30TH AV	LAKE ST	CLASS III	2W	
11	Polyline	2851	12751000	TURK ST	ROSSI AV	WILLARD NORTH ST	CLASS II	2W	
12	Polyline	2834	6286000	GOLDEN GATE AVE	CHABOT TER	TEMASCALTER	CLASS II	2W	
13	Polyline	1526	8542000	MALCON AV	MC ALLISTER ST	FULTON ST	CLASS IV	2W	GREEN PAINT

Task2



The screenshot shows the QGIS 3.16.0 interface. The main map displays a street network in San Francisco. The 'bike_routes' layer is selected in the 'Contents' panel. The 'Catalog' panel is open on the right, showing a list of projects and a search bar. The 'Fields' panel at the bottom shows the attributes of the selected features, including 'id', 'shape', 'objid', 'cns', 'streetname', 'from_id', 'to_id', 'direct', 'surface', and 'barrier'.

ID	Shape	objid	cns	streetname	from_id	to_id	direct	surface	barrier
1	Polyline	3421	8012000	LAKE ST	47th AV	5TH AV	CLASS II	2W	
2	Polyline	3567	8023000	LAKE ST	12TH AV	FULTON AV	CLASS II	2W	
3	Polyline	3620	2322000	ANZA ST	6TH AV	5TH AV	CLASS II	2W	
4	Polyline	1401	8016000	LAKE ST	8TH AV	9TH AV	CLASS II	2W	
5	Polyline	9619	2321000	ANZA ST	5TH AV	4TH AV	CLASS II	2W	
6	Polyline	5005	5123000	EL CAMINO DEL MAR	27TH AV	28TH AV	CLASS II	2W	
7	Polyline	9608	2310000	ANZA ST	SPRUCE ST	COOK ST	CLASS II	2W	
8	Polyline	2008	8018000	LAKE ST	10TH AV	11TH AV	CLASS II	2W	
9	Polyline	4033	12742000	TURK ST	ANNAPOLIS TERR	TAMALPAIS TERR	CLASS II	2W	
10	Polyline	2190	5128000	EL CAMINO DEL MAR	30TH AV	LAKE ST	CLASS III	2W	
11	Polyline	2851	12751000	TURK ST	ROSSI AV	WILLARD NORTH ST	CLASS II	2W	
12	Polyline	2834	6286000	GOLDEN GATE AVE	CHABOT TER	TEMASCALTER	CLASS II	2W	
13	Polyline	1526	8542000	MALCON AV	MC ALLISTER ST	FULTON ST	CLASS IV	2W	GREEN PAINT